For Examiner's Use

4 **(a)** Fig.4.1 shows a string stretched between two fixed points P and Q.

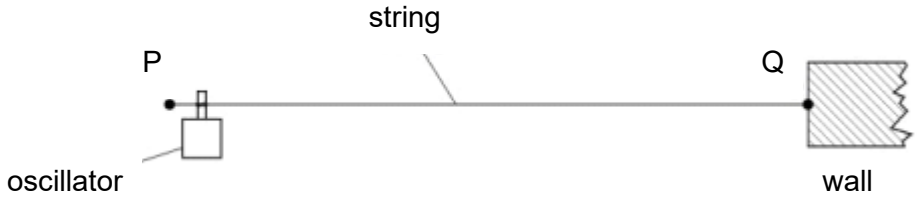

Fig. 4.1

An oscillator is attached near end P of the string. End Q is fixed to a wall. The oscillator has a frequency of 50.0 Hz.

The stationary wave produced on PQ at an instant time *t* is shown in Fig. 4.2. Each point on the string is at its maximum displacement.

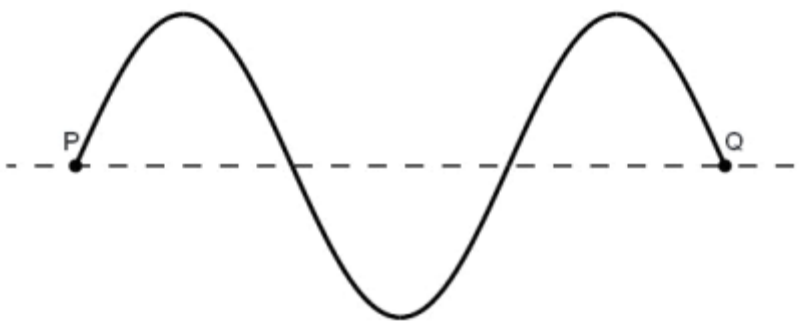

Fig. 4.2

(i) On Fig. 4.2, label all the nodes with the letter **N** and the antinodes with the letter **A** along the dotted line PQ. [1]

(ii) On Fig 4.2, draw the stationary wave at (t + 5.0 ms). [1]

Examiner's Use

(b) Sound waves is directed from a loudspeaker towards a metal plate. A microphone, connected to a cathode ray oscilloscope (CRO), is placed in between the loudspeaker and the metal plate as shown in Fig. 4.3.

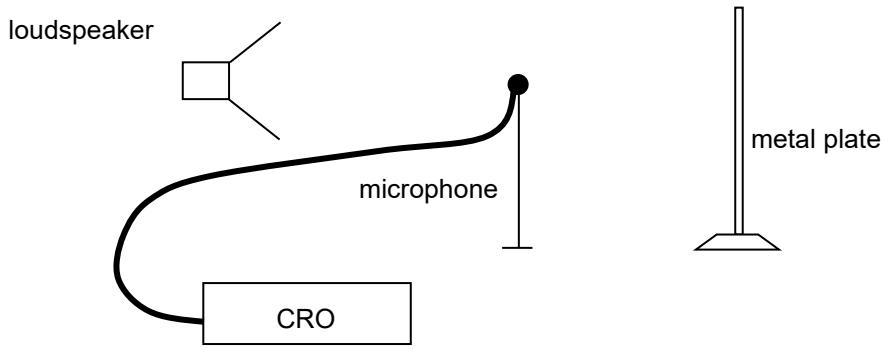

Fig. 4.3

(i) Explain how stationary waves are formed in between the loudspeaker and the metal phase.

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……..[3]

(ii) The trace on the CRO in Fig. 4.4 shows the variation of signal with time at an antinode.

The time-base setting on the CRO is 0.10 ms cm^{-1}.

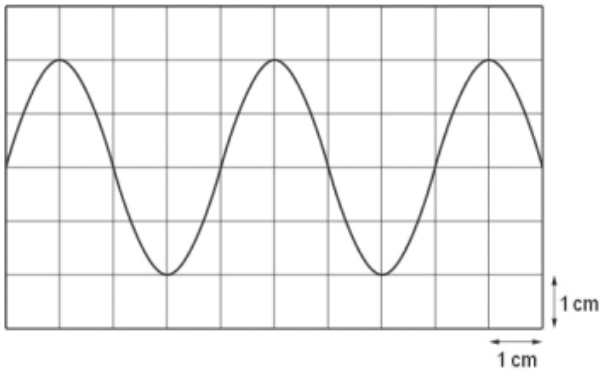

Fig. 4.4

The microphone is then moved by 3.3 cm and the trace on the CRO now records zero amplitude.

1. Determine the frequency of the sound.

frequency = …….. Hz [1]

2. Calculate the speed of sound.

speed = …….. m s^{-1} [2]